ECS 191: Building products

Product Engineering

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Administrative



HW2 Due tomorrow

Individual meetings on Thursday (will post sched.)

- I'll want to see your "core loop" for sure

HW3 includes getting 5 people to use your app

- Today we'll show you what you need to do
- Install app on their phone, go through onboarding + core loop
- Record the interviews, submit recordings
- Failure mode: ignoring the content from today's lecture!

Limited scholarships available for Apple Developer License

Update to attendance policy

Getting people to use your app for the homework

1. **Sideloading**. Laptop + cable, install using Xcode
2. **TestFlight private beta.** Users have to sign up for an account get app from Apple
3. **TestFlight public beta.** Lightweight review from apple + URL. Users need the TestFlight app
4. **App Store.** Full review, install using App Store

First entrepreneurship lecture is today!

But really, it’s all about humans in this class

Complexity is a UX bug, not a set of features!

In this class, focus on making one key feature the best you can, the rest of your app should be spartan.

Focus areas:

- Speed – try out as many features or iterations on the same feature as possible!
- Human feedback – you want to learn something from people that use it at this point
- Insight – get to know the problem you're working on better than anyone else in the world

This is also important for your grade, when you demo your product you'll demo that one key feature not everything

Two phrases I never want to hear from engineers on my team

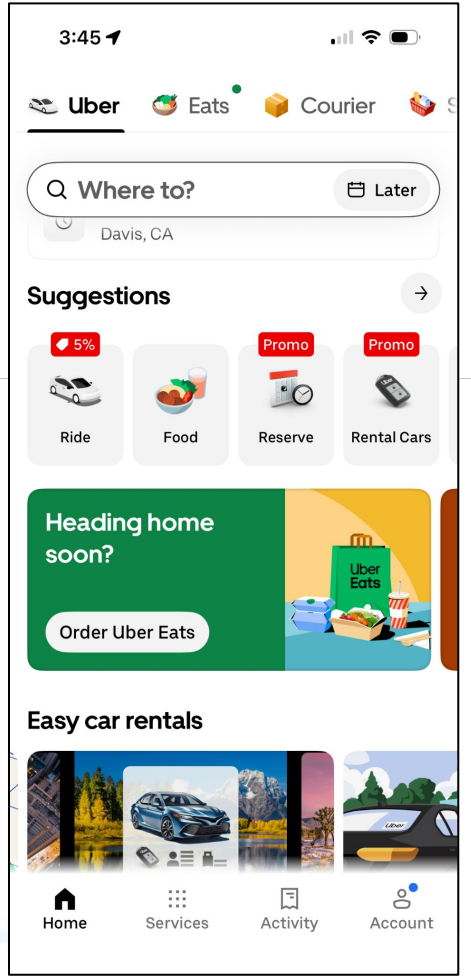
1. That’s a product problem
2. I need a pixel perfect design

The swiss army knife vs the scalpel product





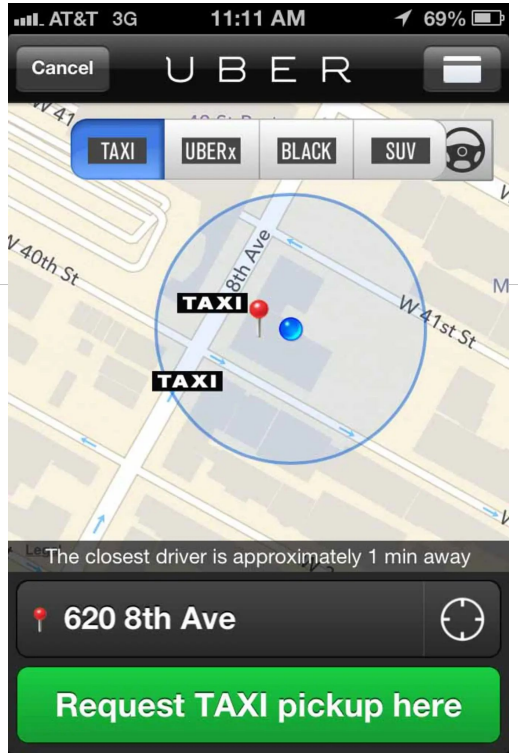
Swiss army knife or scalpel?



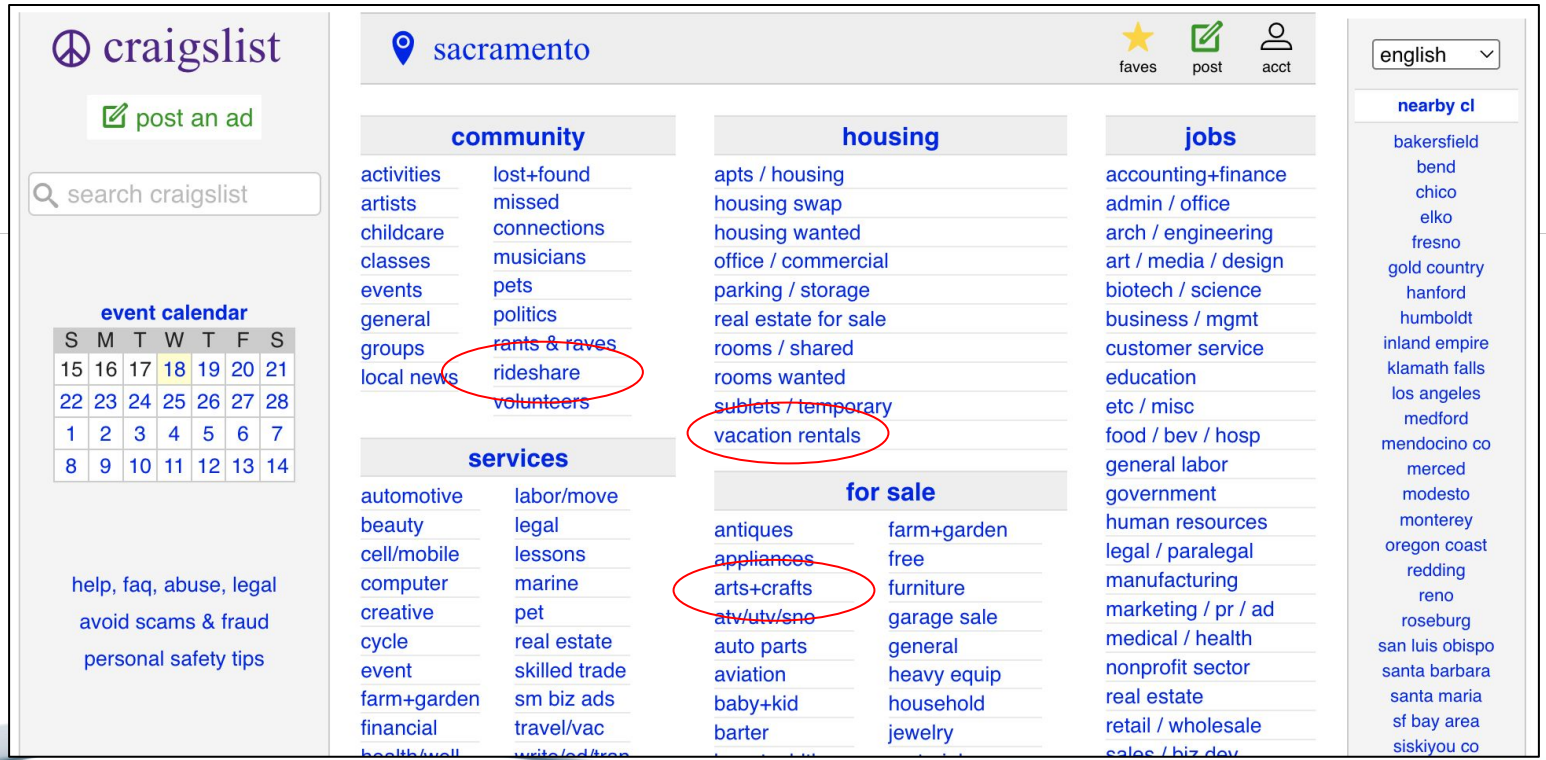
Swiss army knife or scalpel?

Magic of old school Uber

- GPS used in a non maps app
- Hailing a taxi was hard
- Payments handled

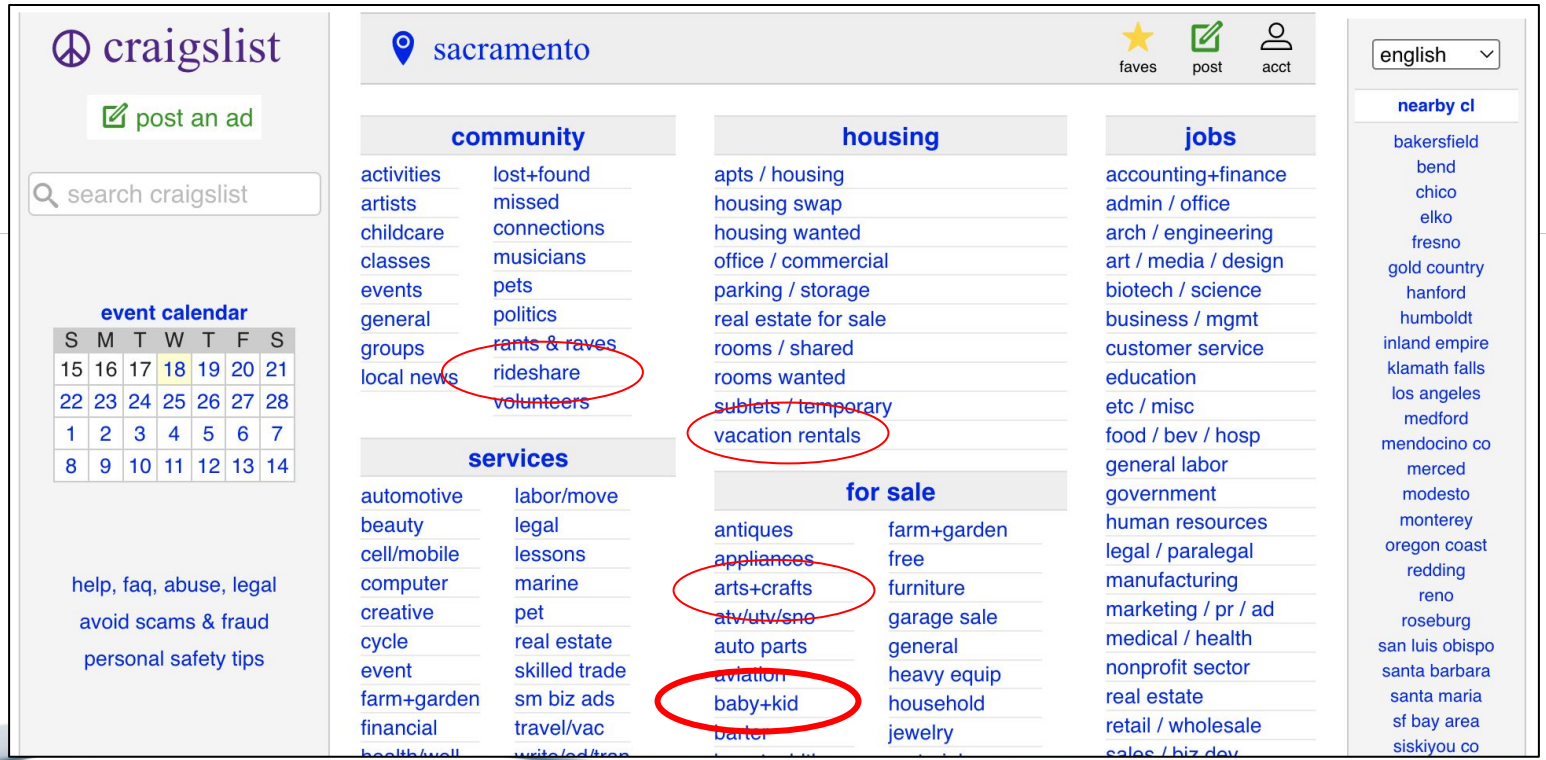


Swiss army knife or scalpel?



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Swiss army knife or scalpel?



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Swiss army knife or scalpel?







Burbn app (aka Instagram)



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Feature creep: Your app is doing too much!

10/10/10 rule. It takes 10 seconds for a user to judge the UI, 10 seconds to understand the value, and 10 seconds to get the first result.

If your app requires a tutorial, you've already lost.

If your app has advanced settings, most people won’t use them

The Challenge: You need to decide what to hide, organize, or delete.

Goal: Reduce the “time to magic”

How fast can a new user get your product to do a “cool thing”?

Identify the core loop for the most important feature

- Some input -> processing (maybe AI) -> some output

You want to make everything obvious

Running simple user experiments

Three main rules:

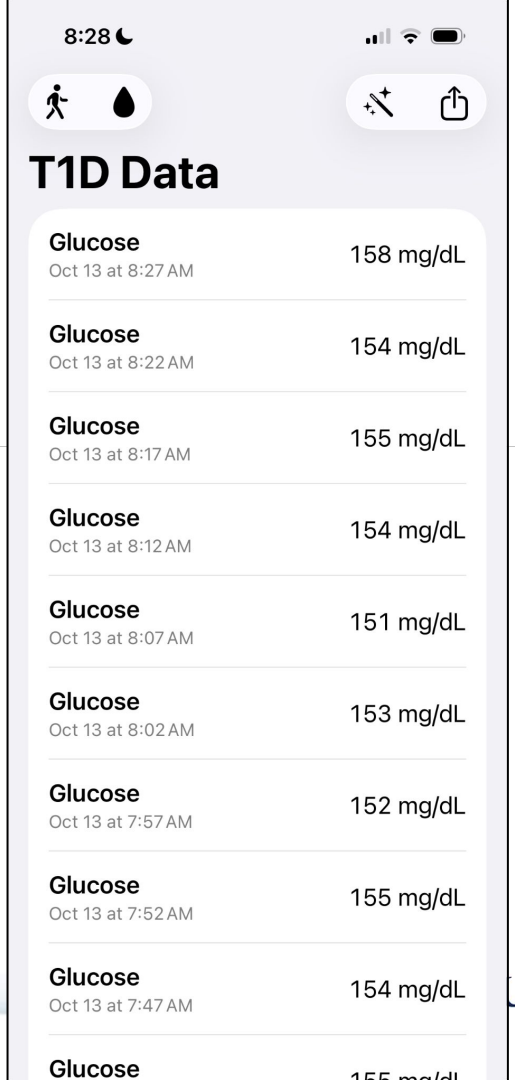
- **Shut Up.** You are observing, not selling.
- **Tasks, Not Tours.** Don't give a tour. Give them a goal ("Book a flight") and observe.
- **Watch the struggle.** The exact moment they get confused is the exact place you need to simplify your code.

The "Show Me" Script

Goal: Focus on "Show me..."

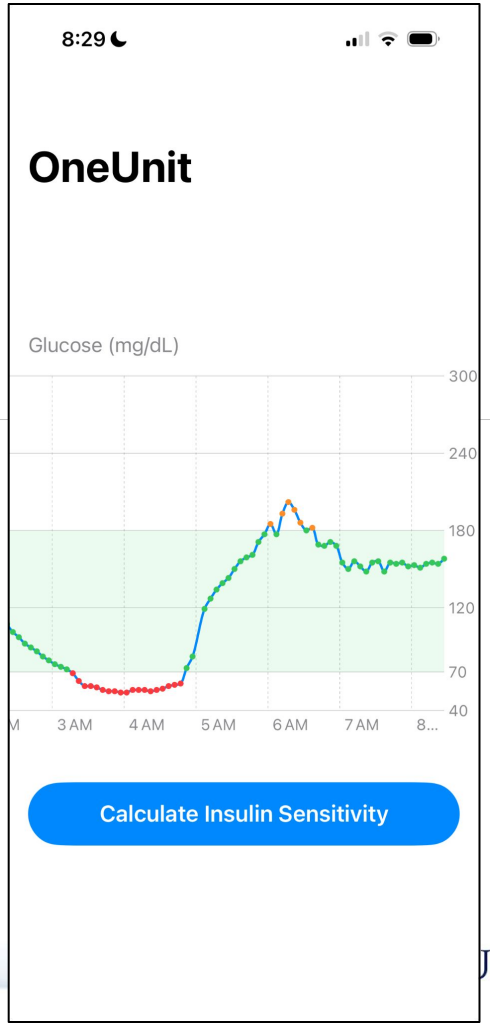
Two volunteers to test out *my* product

Take 20 seconds to look at this screenshot, what does this app do and how do you do it?



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Take 20 seconds to look at this screenshot, what does this app do and how do you do it?



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Bring in the volunteers

Key takeaways

No distractions – only the essentials

- Other features are reasonable, not here

Blatantly obvious what the app is doing

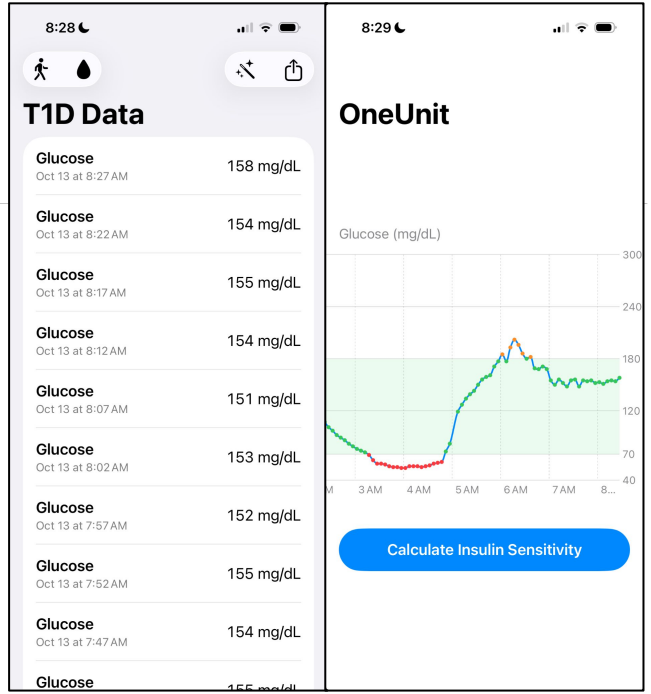
- Explicitly did not automate calculation to provide context

Familiar UI/UX to reduce the cognitive load

Focused on building trust in the AI results

- Accuracy listed, signaling that I'm using their actual data, etc

I explicitly did *not* give my title the phrase AI



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Phase 1: The Setup (lower the stakes)

"I’m going to show you a rough prototype we’re working on. I want to be clear: **I am testing the app, I am not testing you.** If you get stuck or confused, that is great – it helps me fix the design. Please be honest; you won’t hurt my feelings."

Phase 2: The "First Glance" (The 5-Second Test)

- Hand them the device on the home screen. Do NOT give them a task yet.

"Take a look at this screen for a few seconds. Without clicking anything yet... what do you think this app does?"

- Why this matters: If they say "It looks like a calendar" and you built a diet tracker, you have failed before they even started.

Phase 3: The Task (The Core Loop)

Give them a scenario, not instructions.

Script: "Imagine you are planning a trip for next weekend. Show me how you would use this tool to get a list of restaurants."

Silently pay attention to how they use the app.

Confusion is where you need to focus your attention

Phase 4: The AI "Vibe Check"

You need to test the quality of the AI response, not just the buttons.

- "Look at the answer the AI just gave you."
- "Is this useful?"
- "Do you trust it?"
- "What would you do with this information next?"

Phase 5: The Debrief

"On a scale of 1 to 5, how difficult was that?"

"If you had a magic wand, what is the one thing you would change to make this easier?"

"How would you describe this app to a friend in one sentence?"

In class work: Reduce the “time to magic”

How fast can a new user get your product to do a “cool thing”?

Task

- Identify your core loops and pick the most important one
- Create screenshots of this loop from your product
- Gray out everything not related to the core loop (secondary features, “nice to have” aspects, etc)
- Keep both the “before” and “after” screenshots